

M16 / ASSESSMENT

M16 Module Revision Test

Standalone Zero-to-Job-Ready Data Engineer 2026

Study, practise, check and explain.

A complete learning route

Start with the lesson video. Read the worked explanation beside it, pause to reproduce the example, then solve the supplied task. The books contain your study notes; keep your own work and results as evidence.

Use the protected answers after saving your first attempt. A changed retry shows what you can now do independently. There is no forced timer during learning.

One course. M00-M19. 264 lessons and 50 project phases.

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Tap a topic or use the PDF bookmarks. These numbers are the actual pages in this edition.

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Read this first

Rules: Close learner/review/answers. Save your genuine attempt. Untimed mastery first; optional interview timing may be added later. Open protected answers only after the attempt.

1. DE16-01 - Production mindset

Explain why 'the DAG is green' is not enough to call a data product healthy. Give two examples of successful execution with wrong business output.

Then write one failure mode, one validation proof, and one recovery action for the same competency.

2. DE16-02 - Testing strategy for data engineering

Using one orders example, distinguish a unit test, a data-quality expectation, an integration test and a reconciliation check.

Then write one failure mode, one validation proof, and one recovery action for the same competency.

3. DE16-03 - Automated quality gates

Why is a validation report alone not a quality gate? Name the additional control that makes it a gate.

Then write one failure mode, one validation proof, and one recovery action for the same competency.

4. DE16-04 - Continuous Integration fundamentals

Explain CI to someone who believes it means 'automatically deploy every commit.' What is missing from that definition?

Then write one failure mode, one validation proof, and one recovery action for the same competency.

5. DE16-05 - GitHub Actions for data projects

What is the difference between `uses:` and `run:`? Why should workflow permissions be designed explicitly?

Then write one failure mode, one validation proof, and one recovery action for the same competency.

6. DE16-06 - Dev, test and prod environments

Why should promotion prefer the same artifact with different bindings rather than three manually edited copies?

Then write one failure mode, one validation proof, and one recovery action for the same competency.

7. DE16-07 - Deployment and rollback

Why is code rollback not the same as data rollback? Give one example where the old code is restored but the system remains wrong.

Then write one failure mode, one validation proof, and one recovery action for the same competency.

8. DE16-08 - Secrets and credential safety

Why does deleting a leaked credential from the latest source file not make the credential safe again?

Then write one failure mode, one validation proof, and one recovery action for the same competency.

9. DE16-09 - RBAC and least privilege

Explain principal + role + scope with one Azure example. Why can a management-plane role differ from data access?

Then write one failure mode, one validation proof, and one recovery action for the same competency.

10. DE16-10 - PII, encryption and masking

Why are encryption and masking different controls? Give one failure that encryption alone does not prevent.

Then write one failure mode, one validation proof, and one recovery action for the same competency.

11. DE16-11 - Metadata, lineage and governance

What can lineage answer that a schema alone cannot? Give one upstream root-cause and one downstream impact question.

Then write one failure mode, one validation proof, and one recovery action for the same competency.

12. DE16-12 - Observability for data pipelines

Why can CPU/runtime/success metrics all be healthy while a data product is unhealthy? Name three data-specific signals.

Then write one failure mode, one validation proof, and one recovery action for the same competency.

13. DE16-13 - Incidents, runbooks and root cause

What is the difference between a runbook and an RCA? Why should facts and hypotheses be labeled separately during an incident?

Then write one failure mode, one validation proof, and one recovery action for the same competency.

14. DE16-14 - Terraform and IaC foundations

Explain why terraform plan is valuable but does not prove application or data correctness. What extra evidence belongs after infrastructure deployment?

Then write one failure mode, one validation proof, and one recovery action for the same competency.